

# Model for “C3v”

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## General Condition

- Basis type: **1g**
- SAMB selection:
  - Type: **[Q, G]**
  - Rank: **[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]**
  - Irrep.: **[A<sub>1</sub>, A<sub>2</sub>, E]**
  - Spin (s): **[0, 1]**
- Atomic selection:
  - Type: **[Q, G, M, T]**
  - Rank: **[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]**
  - Irrep.: **[A<sub>1</sub>, A<sub>2</sub>, E]**
  - Spin (s): **[0, 1]**
- Site-cluster selection:
  - Rank: **[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]**
  - Irrep.: **[A<sub>1</sub>, A<sub>2</sub>, E]**
- Bond-cluster selection:
  - Type: **[Q, G, M, T]**
  - Rank: **[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]**
  - Irrep.: **[A<sub>1</sub>, A<sub>2</sub>, E]**
- Max. neighbor: **10**
- Search cell range: **(-2, 3), (-2, 3), (-2, 3)**
- Toroidal priority: **false**

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## Group and Unit Cell

- Group: PG No. 35  $C_{3v}(1)$   $3m$  (31m setting) [ trigonal ]
- Unit cell:
  - $a = 1.00000$ ,  $b = 1.00000$ ,  $c = 10.00000$ ,  $\alpha = 90.0$ ,  $\beta = 90.0$ ,  $\gamma = 120.0$
- Lattice vectors (conventional cell):
  - $\mathbf{a}_1 = [ 1.00000, 0.00000, 0.00000 ]$
  - $\mathbf{a}_2 = [ -0.50000, 0.86603, 0.00000 ]$
  - $\mathbf{a}_3 = [ 0.00000, 0.00000, 10.00000 ]$

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## Symmetry Operation

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Table 1: Symmetry operation

| # | SO        | # | SO          | # | SO          | # | SO         | # | SO        |
|---|-----------|---|-------------|---|-------------|---|------------|---|-----------|
| 1 | 1         | 2 | $3_{001}^+$ | 3 | $3_{001}^-$ | 4 | $m_{1-10}$ | 5 | $m_{120}$ |
| 6 | $m_{210}$ |   |             |   |             |   |            |   |           |

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## Harmonics

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Table 2: Harmonics

| # | symbol                 | irrep. | rank | X      | multiplicity | component | symmetry                               |
|---|------------------------|--------|------|--------|--------------|-----------|----------------------------------------|
| 1 | $\mathbb{Q}_0(A_1)$    | $A_1$  | 0    | $Q, T$ | -            | -         | 1                                      |
| 2 | $\mathbb{Q}_1(A_1)$    | $A_1$  | 1    | $Q, T$ | -            | -         | $z$                                    |
| 3 | $\mathbb{Q}_2(A_1)$    | $A_1$  | 2    | $Q, T$ | -            | -         | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| 4 | $\mathbb{Q}_3(A_1, 2)$ | $A_1$  | 3    | $Q, T$ | 2            | -         | $\frac{\sqrt{10}x(x^2-3y^2)}{4}$       |
| 5 | $\mathbb{G}_1(A_2)$    | $A_2$  | 1    | $G, M$ | -            | -         | $z$                                    |
| 6 | $\mathbb{G}_2(A_2)$    | $A_2$  | 2    | $G, M$ | -            | -         | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| 7 | $\mathbb{Q}_3(A_2)$    | $A_2$  | 3    | $Q, T$ | -            | -         | $\frac{\sqrt{10}y(3x^2-y^2)}{4}$       |
| 8 | $\mathbb{Q}_4(A_2)$    | $A_2$  | 4    | $Q, T$ | -            | -         | $\frac{\sqrt{70}yz(3x^2-y^2)}{4}$      |
| 9 | $\mathbb{G}_{1,1}(E)$  | $E$    | 1    | $G, M$ | -            | 1         | $-y$                                   |

*continued ...*

Table 2

| #  | symbol                   | irrep. | rank | X      | multiplicity | component | symmetry                             |
|----|--------------------------|--------|------|--------|--------------|-----------|--------------------------------------|
| 10 | $\mathbb{G}_{1,2}(E)$    |        |      |        |              | 2         | $x$                                  |
| 11 | $\mathbb{Q}_{1,1}(E)$    | $E$    | 1    | $Q, T$ | -            | 1         | $x$                                  |
| 12 | $\mathbb{Q}_{1,2}(E)$    |        |      |        |              | 2         | $y$                                  |
| 13 | $\mathbb{Q}_{2,1}(E, 1)$ | $E$    | 2    | $Q, T$ | 1            | 1         | $\sqrt{3}xz$                         |
| 14 | $\mathbb{Q}_{2,2}(E, 1)$ |        |      |        |              | 2         | $\sqrt{3}yz$                         |
| 15 | $\mathbb{Q}_{2,1}(E, 2)$ | $E$    | 2    | $Q, T$ | 2            | 1         | $\frac{\sqrt{3}(x^2-y^2)}{2}$        |
| 16 | $\mathbb{Q}_{2,2}(E, 2)$ |        |      |        |              | 2         | $-\sqrt{3}xy$                        |
| 17 | $\mathbb{Q}_{3,1}(E, 1)$ | $E$    | 3    | $Q, T$ | 1            | 1         | $\frac{\sqrt{6}x(-x^2-y^2+4z^2)}{4}$ |
| 18 | $\mathbb{Q}_{3,2}(E, 1)$ |        |      |        |              | 2         | $\frac{\sqrt{6}y(-x^2-y^2+4z^2)}{4}$ |
| 19 | $\mathbb{Q}_{3,1}(E, 2)$ | $E$    | 3    | $Q, T$ | 2            | 1         | $\frac{\sqrt{15}z(-x^2+y^2)}{2}$     |
| 20 | $\mathbb{Q}_{3,2}(E, 2)$ |        |      |        |              | 2         | $\sqrt{15}xyz$                       |

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**Basis in full matrix**


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Table 3: dimension = 12

| #  | orbital@atom(SL)    | #  | orbital@atom(SL)    | # | orbital@atom(SL)    | # | orbital@atom(SL)    | # | orbital@atom(SL)    |
|----|---------------------|----|---------------------|---|---------------------|---|---------------------|---|---------------------|
| 0  | $ s\rangle @A(1)$   | 1  | $ s\rangle @A(2)$   | 2 | $ s\rangle @A(3)$   | 3 | $ p_x\rangle @B(1)$ | 4 | $ p_y\rangle @B(1)$ |
| 5  | $ p_z\rangle @B(1)$ | 6  | $ p_x\rangle @B(2)$ | 7 | $ p_y\rangle @B(2)$ | 8 | $ p_z\rangle @B(2)$ | 9 | $ p_x\rangle @B(3)$ |
| 10 | $ p_y\rangle @B(3)$ | 11 | $ p_z\rangle @B(3)$ |   |                     |   |                     |   |                     |

Table 4: Atomic basis (orbital part only)

| orbital       | definition |
|---------------|------------|
| $ s\rangle$   | 1          |
| $ p_x\rangle$ | $x$        |
| $ p_y\rangle$ | $y$        |
| $ p_z\rangle$ | $z$        |

— SAMB: 42 (all 42) —

• **A** : 'A' site-cluster

\* bra:  $\langle s|$

\* ket:  $|s\rangle$

\* wyckoff: **3b**

$$\boxed{\text{z1}} \quad \mathbb{Q}_0^{(c)}(A_1) = \mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_0^{(s)}(A_1)$$

$$\boxed{\text{z15}} \quad \mathbb{Q}_{1,1}^{(c)}(E) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_{1,1}^{(s)}(E)}{2}$$

$$\boxed{\text{z16}} \quad \mathbb{Q}_{1,2}^{(c)}(E) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_{1,2}^{(s)}(E)}{2}$$

• **B** : 'B' site-cluster

\* bra:  $\langle p_x|, \langle p_y|, \langle p_z|$

\* ket:  $|p_x\rangle, |p_y\rangle, |p_z\rangle$

\* wyckoff: **3b**

$$\boxed{\text{z2}} \quad \mathbb{Q}_0^{(c)}(A_1) = \mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_0^{(s)}(A_1)$$

$$\boxed{\text{z3}} \quad \mathbb{Q}_1^{(c)}(A_1) = \frac{\sqrt{2} \left( \mathbb{Q}_{2,1}^{(a)}(E, 1)\mathbb{Q}_{1,1}^{(s)}(E) + \mathbb{Q}_{2,2}^{(a)}(E, 1)\mathbb{Q}_{1,2}^{(s)}(E) \right)}{2}$$

$$\boxed{\text{z4}} \quad \mathbb{Q}_2^{(c)}(A_1) = \mathbb{Q}_2^{(a)}(A_1)\mathbb{Q}_0^{(s)}(A_1)$$

$$\boxed{\text{z5}} \quad \mathbb{Q}_3^{(c)}(A_1, 2) = \frac{\sqrt{2} \left( \mathbb{Q}_{2,1}^{(a)}(E, 2)\mathbb{Q}_{1,1}^{(s)}(E) + \mathbb{Q}_{2,2}^{(a)}(E, 2)\mathbb{Q}_{1,2}^{(s)}(E) \right)}{2}$$

$$\boxed{\text{z10}} \quad \mathbb{Q}_3^{(c)}(A_2) = \frac{\sqrt{2} \left( \mathbb{Q}_{2,1}^{(a)}(E, 2)\mathbb{Q}_{1,2}^{(s)}(E) - \mathbb{Q}_{2,2}^{(a)}(E, 2)\mathbb{Q}_{1,1}^{(s)}(E) \right)}{2}$$

$$\boxed{\text{z11}} \quad \mathbb{G}_2^{(c)}(A_2) = \frac{\sqrt{2} \left( \mathbb{Q}_{2,1}^{(a)}(E, 1)\mathbb{Q}_{1,2}^{(s)}(E) - \mathbb{Q}_{2,2}^{(a)}(E, 1)\mathbb{Q}_{1,1}^{(s)}(E) \right)}{2}$$

$$\boxed{\text{z17}} \quad \mathbb{Q}_{1,1}^{(c)}(E, a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_{1,1}^{(s)}(E)}{2}$$

$$\boxed{\text{z18}} \quad \mathbb{Q}_{1,2}^{(c)}(E, a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_{1,2}^{(s)}(E)}{2}$$

$$\boxed{\text{z19}} \quad \mathbb{Q}_{1,1}^{(c)}(E, b) = \frac{\sqrt{14} \left( \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E, 2)\mathbb{Q}_{1,1}^{(s)}(E) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E, 2)\mathbb{Q}_{1,2}^{(s)}(E) - \mathbb{Q}_2^{(a)}(A_1)\mathbb{Q}_{1,1}^{(s)}(E) \right)}{14}$$

$$\boxed{\text{z20}} \quad \mathbb{Q}_{1,2}^{(c)}(E, b) = -\frac{\sqrt{14} \left( \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E, 2)\mathbb{Q}_{1,2}^{(s)}(E) + \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E, 2)\mathbb{Q}_{1,1}^{(s)}(E) + \mathbb{Q}_2^{(a)}(A_1)\mathbb{Q}_{1,2}^{(s)}(E) \right)}{14}$$

$$\boxed{\text{z21}} \quad \mathbb{Q}_{2,1}^{(c)}(E, 1) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E, 1)\mathbb{Q}_0^{(s)}(A_1)}{2}$$

$$\boxed{\text{z22}} \quad \mathbb{Q}_{2,2}^{(c)}(E, 1) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E, 1)\mathbb{Q}_0^{(s)}(A_1)}{2}$$

$$\boxed{\text{z23}} \quad \mathbb{Q}_{2,1}^{(c)}(E, 2) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E, 2)\mathbb{Q}_0^{(s)}(A_1)}{2}$$

$$\boxed{\text{z24}} \quad \mathbb{Q}_{2,2}^{(c)}(E, 2) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E, 2)\mathbb{Q}_0^{(s)}(A_1)}{2}$$

$$\boxed{\text{z25}} \quad \mathbb{Q}_{3,1}^{(c)}(E, 1) = \frac{\sqrt{7} \left( \mathbb{Q}_{2,1}^{(a)}(E, 2)\mathbb{Q}_{1,1}^{(s)}(E) - \mathbb{Q}_{2,2}^{(a)}(E, 2)\mathbb{Q}_{1,2}^{(s)}(E) + 2\sqrt{3}\mathbb{Q}_2^{(a)}(A_1)\mathbb{Q}_{1,1}^{(s)}(E) \right)}{14}$$

$$\boxed{\text{z26}} \quad \mathbb{Q}_{3,2}^{(c)}(E, 1) = \frac{\sqrt{7} \left( -\mathbb{Q}_{2,1}^{(a)}(E, 2)\mathbb{Q}_{1,2}^{(s)}(E) - \mathbb{Q}_{2,2}^{(a)}(E, 2)\mathbb{Q}_{1,1}^{(s)}(E) + 2\sqrt{3}\mathbb{Q}_2^{(a)}(A_1)\mathbb{Q}_{1,2}^{(s)}(E) \right)}{14}$$

$$\boxed{\text{z27}} \quad \mathbb{Q}_{3,1}^{(c)}(E, 2) = \frac{-\mathbb{Q}_{2,1}^{(a)}(E, 1)\mathbb{Q}_{1,1}^{(s)}(E) + \mathbb{Q}_{2,2}^{(a)}(E, 1)\mathbb{Q}_{1,2}^{(s)}(E)}{2}$$

$$\boxed{\text{z28}} \quad \mathbb{Q}_{3,2}^{(c)}(E, 2) = \frac{\mathbb{Q}_{2,1}^{(a)}(E, 1)\mathbb{Q}_{1,2}^{(s)}(E) + \mathbb{Q}_{2,2}^{(a)}(E, 1)\mathbb{Q}_{1,1}^{(s)}(E)}{2}$$

• **A;A\_001\_1** : 'A'-'A' bond-cluster

\* bra:  $\langle s |$

\* ket:  $|s\rangle$

\* wyckoff: **3b@3b**

$$\boxed{\text{z6}} \quad \mathbb{Q}_0^{(c)}(A_1) = \mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_0^{(b)}(A_1)$$

$$\boxed{\text{z29}} \quad \mathbb{Q}_{1,1}^{(c)}(E) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$$

$$\boxed{\text{z30}} \quad \mathbb{Q}_{1,2}^{(c)}(E) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$$

• **B;A\_001\_1** : 'A'-'B' bond-cluster

\* bra:  $\langle s |$

\* ket:  $|p_x\rangle, |p_y\rangle, |p_z\rangle$

\* wyckoff: **6a@6c**

$$\boxed{\text{z7}} \quad \mathbb{Q}_0^{(c)}(A_1) = \frac{\sqrt{2} \left( \mathbb{Q}_{1,1}^{(a)}(E)\mathbb{Q}_{1,1}^{(b)}(E) + \mathbb{Q}_{1,2}^{(a)}(E)\mathbb{Q}_{1,2}^{(b)}(E) \right)}{2}$$

$$\boxed{\text{z8}} \quad \mathbb{Q}_1^{(c)}(A_1) = \mathbb{Q}_1^{(a)}(A_1)\mathbb{Q}_0^{(b)}(A_1)$$

$$\boxed{\text{z9}} \quad \mathbb{Q}_3^{(c)}(A_1, 2) = \frac{\sqrt{2} \left( \mathbb{Q}_{1,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E, 2) + \mathbb{Q}_{1,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E, 2) \right)}{2}$$

$$\boxed{\text{z12}} \quad \mathbb{Q}_3^{(c)}(A_2) = \frac{\sqrt{2} \left( -\mathbb{Q}_{1,1}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E, 2) + \mathbb{Q}_{1,2}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E, 2) \right)}{2}$$

$$\boxed{\text{z13}} \quad \mathbb{Q}_4^{(c)}(A_2) = \mathbb{Q}_1^{(a)}(A_1)\mathbb{Q}_3^{(b)}(A_2)$$

$$\boxed{\text{z14}} \quad \mathbb{G}_1^{(c)}(A_2) = \frac{\sqrt{2} \left( \mathbb{Q}_{1,1}^{(a)}(E) \mathbb{Q}_{1,2}^{(b)}(E) - \mathbb{Q}_{1,2}^{(a)}(E) \mathbb{Q}_{1,1}^{(b)}(E) \right)}{2}$$

$$\boxed{\text{z31}} \quad \mathbb{Q}_{1,1}^{(c)}(E, a) = \frac{\sqrt{2} \mathbb{Q}_{1,1}^{(a)}(E) \mathbb{Q}_0^{(b)}(A_1)}{2}$$

$$\boxed{\text{z32}} \quad \mathbb{Q}_{1,2}^{(c)}(E, a) = \frac{\sqrt{2} \mathbb{Q}_{1,2}^{(a)}(E) \mathbb{Q}_0^{(b)}(A_1)}{2}$$

$$\boxed{\text{z33}} \quad \mathbb{Q}_{1,1}^{(c)}(E, b) = \frac{\mathbb{Q}_{1,1}^{(a)}(E) \mathbb{Q}_{2,1}^{(b)}(E, 2) - \mathbb{Q}_{1,2}^{(a)}(E) \mathbb{Q}_{2,2}^{(b)}(E, 2)}{2}$$

$$\boxed{\text{z34}} \quad \mathbb{Q}_{1,2}^{(c)}(E, b) = -\frac{\mathbb{Q}_{1,1}^{(a)}(E) \mathbb{Q}_{2,2}^{(b)}(E, 2) + \mathbb{Q}_{1,2}^{(a)}(E) \mathbb{Q}_{2,1}^{(b)}(E, 2)}{2}$$

$$\boxed{\text{z35}} \quad \mathbb{Q}_{2,1}^{(c)}(E, 1) = \frac{\sqrt{2} \mathbb{Q}_1^{(a)}(A_1) \mathbb{Q}_{1,1}^{(b)}(E)}{2}$$

$$\boxed{\text{z36}} \quad \mathbb{Q}_{2,2}^{(c)}(E, 1) = \frac{\sqrt{2} \mathbb{Q}_1^{(a)}(A_1) \mathbb{Q}_{1,2}^{(b)}(E)}{2}$$

$$\boxed{\text{z37}} \quad \mathbb{Q}_{2,1}^{(c)}(E, 2a) = \frac{\sqrt{2} \mathbb{Q}_{1,2}^{(a)}(E) \mathbb{Q}_3^{(b)}(A_2)}{2}$$

$$\boxed{\text{z38}} \quad \mathbb{Q}_{2,2}^{(c)}(E, 2a) = -\frac{\sqrt{2} \mathbb{Q}_{1,1}^{(a)}(E) \mathbb{Q}_3^{(b)}(A_2)}{2}$$

$$\boxed{\text{z39}} \quad \mathbb{Q}_{2,1}^{(c)}(E, 2b) = \frac{\mathbb{Q}_{1,1}^{(a)}(E) \mathbb{Q}_{1,1}^{(b)}(E) - \mathbb{Q}_{1,2}^{(a)}(E) \mathbb{Q}_{1,2}^{(b)}(E)}{2}$$

$$\boxed{\text{z40}} \quad \mathbb{Q}_{2,2}^{(c)}(E, 2b) = -\frac{\mathbb{Q}_{1,1}^{(a)}(E) \mathbb{Q}_{1,2}^{(b)}(E) + \mathbb{Q}_{1,2}^{(a)}(E) \mathbb{Q}_{1,1}^{(b)}(E)}{2}$$

$$\boxed{\text{z41}} \quad \mathbb{Q}_{3,1}^{(c)}(E, 2) = -\frac{\sqrt{2} \mathbb{Q}_1^{(a)}(A_1) \mathbb{Q}_{2,1}^{(b)}(E, 2)}{2}$$

$$\boxed{\text{z42}} \quad \mathbb{Q}_{3,2}^{(c)}(E, 2) = -\frac{\sqrt{2} \mathbb{Q}_1^{(a)}(A_1) \mathbb{Q}_{2,2}^{(b)}(E, 2)}{2}$$

- bra:  $\langle s|$
- ket:  $|s\rangle$

$$\boxed{\text{x1}} \quad \mathbb{Q}_0^{(a)}(A_1) = \begin{bmatrix} 1 \end{bmatrix}$$

- bra:  $\langle p_x|, \langle p_y|, \langle p_z|$
- ket:  $|p_x\rangle, |p_y\rangle, |p_z\rangle$

$$\boxed{\text{x2}} \quad \mathbb{Q}_0^{(a)}(A_1) = \begin{bmatrix} \frac{\sqrt{3}}{3} & 0 & 0 \\ 0 & \frac{\sqrt{3}}{3} & 0 \\ 0 & 0 & \frac{\sqrt{3}}{3} \end{bmatrix}$$

$$\boxed{\text{x3}} \quad \mathbb{Q}_2^{(a)}(A_1) = \begin{bmatrix} -\frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & \frac{\sqrt{6}}{3} \end{bmatrix}$$

$$\boxed{\text{x4}} \quad \mathbb{Q}_{2,1}^{(a)}(E, 1) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}}{2} \\ 0 & 0 & 0 \\ \frac{\sqrt{2}}{2} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x5}} \quad \mathbb{Q}_{2,2}^{(a)}(E, 1) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}}{2} \\ 0 & \frac{\sqrt{2}}{2} & 0 \end{bmatrix}$$

$$\boxed{\text{x6}} \quad \mathbb{Q}_{2,1}^{(a)}(E, 2) = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x7}} \quad \mathbb{Q}_{2,2}^{(a)}(E, 2) = \begin{bmatrix} 0 & -\frac{\sqrt{2}}{2} & 0 \\ -\frac{\sqrt{2}}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$



$$\boxed{\text{x8}} \quad \mathbb{M}_1^{(a)}(A_2) = \begin{bmatrix} 0 & -\frac{\sqrt{2}i}{2} & 0 \\ \frac{\sqrt{2}i}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x9}} \quad \mathbb{M}_{1,1}^{(a)}(E) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}i}{2} \\ 0 & 0 & 0 \\ \frac{\sqrt{2}i}{2} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x10}} \quad \mathbb{M}_{1,2}^{(a)}(E) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{2} \\ 0 & \frac{\sqrt{2}i}{2} & 0 \end{bmatrix}$$

- bra:  $\langle s|$
- ket:  $|p_x\rangle, |p_y\rangle, |p_z\rangle$

$$\boxed{\text{x11}} \quad \mathbb{Q}_1^{(a)}(A_1) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}}{2} \end{bmatrix}$$

$$\boxed{\text{x12}} \quad \mathbb{Q}_{1,1}^{(a)}(E) = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x13}} \quad \mathbb{Q}_{1,2}^{(a)}(E) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{2} & 0 \end{bmatrix}$$

$$\boxed{\text{x14}} \quad \mathbb{T}_1^{(a)}(A_1) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}i}{2} \end{bmatrix}$$

$$\boxed{\text{x15}} \quad \mathbb{T}_{1,1}^{(a)}(E) = \begin{bmatrix} \frac{\sqrt{2}i}{2} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x16}} \quad \mathbb{T}_{1,2}^{(a)}(E) = \begin{bmatrix} 0 & \frac{\sqrt{2}i}{2} & 0 \end{bmatrix}$$

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## Cluster SAMB

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- Site cluster

\*\* Wyckoff: 3b

$$\boxed{\text{y1}} \quad \mathbb{Q}_0^{(s)}(A_1) = \left[ \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3} \right]$$

$$\boxed{\text{y2}} \quad \mathbb{Q}_{1,1}^{(s)}(E) = \left[ \frac{\sqrt{6}}{3}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6} \right]$$

$$\boxed{\text{y3}} \quad \mathbb{Q}_{1,2}^{(s)}(E) = \left[ 0, \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2} \right]$$

• Bond cluster

\*\* Wyckoff: 3b@3b

$$\boxed{\text{y4}} \quad \mathbb{Q}_0^{(b)}(A_1) = \left[ \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3} \right]$$

$$\boxed{\text{y5}} \quad \mathbb{M}_1^{(b)}(A_2) = \left[ \frac{\sqrt{3}i}{3}, \frac{\sqrt{3}i}{3}, \frac{\sqrt{3}i}{3} \right]$$

$$\boxed{\text{y6}} \quad \mathbb{Q}_{1,1}^{(b)}(E) = \left[ \frac{\sqrt{6}}{3}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6} \right]$$

$$\boxed{\text{y7}} \quad \mathbb{Q}_{1,2}^{(b)}(E) = \left[ 0, \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2} \right]$$

$$\boxed{\text{y8}} \quad \mathbb{T}_{1,1}^{(b)}(E) = \left[ 0, -\frac{\sqrt{2}i}{2}, \frac{\sqrt{2}i}{2} \right]$$

$$\boxed{\text{y9}} \quad \mathbb{T}_{1,2}^{(b)}(E) = \left[ \frac{\sqrt{6}i}{3}, -\frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6} \right]$$

\*\* Wyckoff: 6a@6c

$$\boxed{\text{y10}} \quad \mathbb{Q}_0^{(b)}(A_1) = \left[ \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6} \right]$$

$$\boxed{\text{y11}} \quad \mathbb{T}_0^{(b)}(A_1) = \left[ \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6} \right]$$

$$\boxed{\text{y12}} \quad \mathbb{M}_1^{(b)}(A_2) = \left[ \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6} \right]$$

$$\boxed{\text{y13}} \quad \mathbb{Q}_3^{(b)}(A_2) = \left[ \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6} \right]$$

$$\boxed{\text{y14}} \quad \mathbb{Q}_{1,1}^{(b)}(E) = \left[ \frac{5\sqrt{21}}{42}, -\frac{2\sqrt{21}}{21}, -\frac{\sqrt{21}}{42}, -\frac{\sqrt{21}}{42}, \frac{5\sqrt{21}}{42}, -\frac{2\sqrt{21}}{21} \right]$$

$$\boxed{\text{y15}} \quad \mathbb{Q}_{1,2}^{(b)}(E) = \left[ \frac{\sqrt{7}}{14}, \frac{\sqrt{7}}{7}, -\frac{3\sqrt{7}}{14}, \frac{3\sqrt{7}}{14}, -\frac{\sqrt{7}}{14}, -\frac{\sqrt{7}}{7} \right]$$

$$\boxed{\text{y16}} \quad \mathbb{T}_{1,1}^{(b)}(E, a) = \left[ \frac{5\sqrt{21}i}{42}, -\frac{2\sqrt{21}i}{21}, -\frac{\sqrt{21}i}{42}, -\frac{\sqrt{21}i}{42}, \frac{5\sqrt{21}i}{42}, -\frac{2\sqrt{21}i}{21} \right]$$

$$\boxed{\text{y17}} \quad \mathbb{T}_{1,2}^{(b)}(E, a) = \left[ \frac{\sqrt{7}i}{14}, \frac{\sqrt{7}i}{7}, -\frac{3\sqrt{7}i}{14}, \frac{3\sqrt{7}i}{14}, -\frac{\sqrt{7}i}{14}, -\frac{\sqrt{7}i}{7} \right]$$

$$\boxed{\text{y18}} \quad \mathbb{T}_{1,1}^{(b)}(E, b) = \left[ \frac{\sqrt{7}i}{14}, \frac{\sqrt{7}i}{7}, -\frac{3\sqrt{7}i}{14}, -\frac{3\sqrt{7}i}{14}, \frac{\sqrt{7}i}{14}, \frac{\sqrt{7}i}{7} \right]$$

$$\boxed{\text{y19}} \quad \mathbb{T}_{1,2}^{(b)}(E, b) = \left[ -\frac{5\sqrt{21}i}{42}, \frac{2\sqrt{21}i}{21}, \frac{\sqrt{21}i}{42}, -\frac{\sqrt{21}i}{42}, \frac{5\sqrt{21}i}{42}, -\frac{2\sqrt{21}i}{21} \right]$$

$$\boxed{\text{y20}} \quad \mathbb{Q}_{2,1}^{(b)}(E, 2) = \left[ \frac{\sqrt{7}}{14}, \frac{\sqrt{7}}{7}, -\frac{3\sqrt{7}}{14}, -\frac{3\sqrt{7}}{14}, \frac{\sqrt{7}}{14}, \frac{\sqrt{7}}{7} \right]$$

$$\boxed{\text{y21}} \quad \mathbb{Q}_{2,2}^{(b)}(E, 2) = \left[ -\frac{5\sqrt{21}}{42}, \frac{2\sqrt{21}}{21}, \frac{\sqrt{21}}{42}, -\frac{\sqrt{21}}{42}, \frac{5\sqrt{21}}{42}, -\frac{2\sqrt{21}}{21} \right]$$

— Site and Bond —

Table 5: Orbital of each site

| # | site | orbital                                 |
|---|------|-----------------------------------------|
| 1 | A    | $ s\rangle$                             |
| 2 | B    | $ p_x\rangle,  p_y\rangle,  p_z\rangle$ |

Table 6: Neighbor and bra-ket of each bond

| # | head | tail | neighbor | head (bra) | tail (ket) |
|---|------|------|----------|------------|------------|
| 1 | A    | A    | [1]      | [s]        | [s]        |
| 2 | A    | B    | [1]      | [s]        | [p]        |

### — Site in Unit Cell —

Sites in (conventional) cell (no plus set), SL = sublattice

Table 7: 'A' (#1) site cluster (3b), . . m

| SL | position ( $s$ )             | mapping |
|----|------------------------------|---------|
| 1  | [ 0.16667, 0.00000, 0.00000] | [1,5]   |
| 2  | [ 0.00000, 0.16667, 0.00000] | [2,4]   |
| 3  | [-0.16667,-0.16667, 0.00000] | [3,6]   |

Table 8: 'B' (#2) site cluster (3b), . . m

| SL | position ( $\mathbf{s}$ )    | mapping |
|----|------------------------------|---------|
| 1  | [-0.66667, 0.00000, 0.00000] | [1,5]   |
| 2  | [ 0.00000,-0.66667, 0.00000] | [2,4]   |
| 3  | [ 0.66667, 0.66667, 0.00000] | [3,6]   |

### Bond in Unit Cell

Bonds in (conventional) cell (no plus set): tail, head = (SL, plus set), (N)D = (non)directional (listed up to 5th neighbor at most)

Table 9: 1-th 'A'-'A' [1] (#1) bond cluster (3b@3b), ND,  $|\mathbf{v}| = 0.28868$  (cartesian)

| SL | vector ( $\mathbf{v}$ )      | center ( $\mathbf{c}$ )      | mapping | head  | tail  | $\mathbf{R}$ (primitive) |
|----|------------------------------|------------------------------|---------|-------|-------|--------------------------|
| 1  | [-0.16667,-0.33333, 0.00000] | [-0.08333, 0.00000, 0.00000] | [1,-5]  | (3,1) | (2,1) | [0,0,0]                  |
| 2  | [ 0.33333, 0.16667, 0.00000] | [ 0.00000,-0.08333, 0.00000] | [2,-4]  | (1,1) | (3,1) | [0,0,0]                  |
| 3  | [-0.16667, 0.16667, 0.00000] | [ 0.08333, 0.08333, 0.00000] | [3,-6]  | (2,1) | (1,1) | [0,0,0]                  |

Table 10: 1-th 'A'-'B' [1] (#2) bond cluster (6a@6c), D,  $|\mathbf{v}| = 0.60093$  (cartesian)

| SL | vector ( $\mathbf{v}$ )      | center ( $\mathbf{c}$ )      | mapping | head  | tail  | $\mathbf{R}$ (primitive) |
|----|------------------------------|------------------------------|---------|-------|-------|--------------------------|
| 1  | [ 0.66667, 0.16667, 0.00000] | [-0.33333, 0.08333, 0.00000] | [1]     | (2,1) | (1,1) | [0,0,0]                  |

*continued ...*

Table 10

| SL | vector ( $\boldsymbol{v}$ )     | center ( $\boldsymbol{c}$ )     | mapping | head                   | tail                   | $\boldsymbol{R}$ (primitive) |
|----|---------------------------------|---------------------------------|---------|------------------------|------------------------|------------------------------|
| 2  | $[-0.16667, 0.50000, 0.00000]$  | $[-0.08333, -0.41667, 0.00000]$ | [2]     | $\langle 3, 1 \rangle$ | $\langle 2, 1 \rangle$ | $[0, 0, 0]$                  |
| 3  | $[-0.50000, -0.66667, 0.00000]$ | $[0.41667, 0.33333, 0.00000]$   | [3]     | $\langle 1, 1 \rangle$ | $\langle 3, 1 \rangle$ | $[0, 0, 0]$                  |
| 4  | $[0.16667, 0.66667, 0.00000]$   | $[0.08333, -0.33333, 0.00000]$  | [4]     | $\langle 1, 1 \rangle$ | $\langle 2, 1 \rangle$ | $[0, 0, 0]$                  |
| 5  | $[0.50000, -0.16667, 0.00000]$  | $[-0.41667, -0.08333, 0.00000]$ | [5]     | $\langle 3, 1 \rangle$ | $\langle 1, 1 \rangle$ | $[0, 0, 0]$                  |
| 6  | $[-0.66667, -0.50000, 0.00000]$ | $[0.33333, 0.41667, 0.00000]$   | [6]     | $\langle 2, 1 \rangle$ | $\langle 3, 1 \rangle$ | $[0, 0, 0]$                  |